

LUMINARY ACADEMY

Mathematics Workbook

Grade 9 Mathematics

Luminary EdTech Ltd. · 2026

Table of Contents

Chapter 1 — Number Systems & Real Numbers	3
Chapter 2 — Algebraic Expressions & Polynomials	8
Chapter 3 — Linear Equations & Inequalities	13
Chapter 4 — Coordinate Geometry	18
Chapter 5 — Introduction to Statistics	23

Chapter 1

Number Systems & Real Numbers

Learning Objectives

- Classify numbers into natural, whole, integer, rational, and irrational sets
- Perform operations on real numbers and apply order of operations
- Understand absolute value and its applications
- Convert between fractions, decimals, and percentages

Essential Vocabulary

- Natural Numbers (N): $\{1, 2, 3, \dots\}$
- Whole Numbers (W): $\{0, 1, 2, 3, \dots\}$
- Integers (Z): $\{\dots, -2, -1, 0, 1, 2, \dots\}$
- Rational Numbers (Q): numbers expressible as p/q , where $q \neq 0$
- Irrational Numbers: non-repeating, non-terminating decimals (e.g. $\sqrt{2}$, π)
- Real Numbers (R): all rational and irrational numbers
- Absolute Value $|x|$: distance of x from zero on the number line

1.1 Key Formulas & Rules

Order of Operations (PEMDAS/BODMAS):

Parentheses $\rightarrow$ Exponents $\rightarrow$ Multiplication/Division $\rightarrow$ Addition/Subtraction

Absolute Value:

$|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$

Properties of Real Numbers:

- Commutative: $a + b = b + a$ and $a \times b = b \times a$
- Associative: $(a + b) + c = a + (b + c)$

- Distributive: $a(b + c) = ab + ac$
- Identity: $a + 0 = a$ and $a \times 1 = a$
- Inverse: $a + (-a) = 0$ and $a \times (1/a) = 1$ ($a \neq 0$)

1.2 Worked Examples

Example 1

Classify each number: -7 , 0 , $3/4$, $\sqrt{5}$, $2.3\overline{7}$

Solution:

-7 : Integer, Rational, Real 0 : Whole, Integer, Rational, Real $3/4$: Rational, Real $\sqrt{5}$: Irrational, Real $2.3\overline{7} = 7/3$: Rational, Real

Example 2

Evaluate: $3 + 4 \times (2^2 - 1) \div 5$

Solution:

Step 1: Parentheses $\rightarrow 2^2 - 1 = 4 - 1 = 3$ **Step 2: Multiply** $\rightarrow 4 \times 3 = 12$ **Step 3: Divide** $\rightarrow 12 \div 5 = 2.4$ **Step 4: Add** $\rightarrow 3 + 2.4 = 5.4$

Example 3

Evaluate $|-8| + |3 - 7|$

Solution:

$|-8| = 8$ $|3 - 7| = |-4| = 4$ $8 + 4 = 12$

Example 4

Convert 0.625 to a fraction in lowest terms.

Solution:

$0.625 = 625/1000 = 5/8 \checkmark$

■ Common Mistakes

- Do NOT apply absolute value by simply removing the minus sign — always evaluate the expression inside first.

• Remember: $\sqrt{4} = 2$, not ± 2 (the principal root is always non-negative).

1.3 Guided Practice

Guided Practice — Work through with your tutor

1. List all sets that contain the number -3 .

Answer: _____

2. Evaluate: $2 \times (3 + 4^2) - 10 \div 2$

Answer: _____

3. Is $\sqrt{9}$ rational or irrational? Explain.

Answer: _____

4. Place on a number line: -2.5 , $1/3$, $\sqrt{2}$, -1

Answer: _____

1.4 Independent Practice

Independent Practice

1. Classify each: $0.333\dots$, -15 , $\sqrt{16}$, π , $7/2$

Answer: _____

2. Evaluate: $|-12| - |4 - 9| + 3^2$

Answer: _____

3. Evaluate using PEMDAS: $5 + 3 \times 2^2 - (8 \div 4)$

Answer: _____

4. Write 0.875 as a fraction in simplest form.

Answer: _____

5. Which is greater: $-3/4$ or -0.8 ? Show your reasoning.

Answer: _____

6. True or False: Every whole number is a natural number.

Answer: _____

7. Find all integers n such that $|n| = 5$.

Answer: _____

8. Simplify: $-(-(-3))$

Answer: _____

1.5 Exit Ticket

Exit Ticket (5 questions — 5 minutes)

1. Name one number that is irrational.

Answer: _____

2. Evaluate: $|-6 + 2|$

Answer: _____

3. What property is shown: $3(x + 4) = 3x + 12$?

Answer: _____

4. Is 0 a natural number? Yes or No.

Answer: _____

5. Evaluate: $(2 + 3)^2 - 4 \times 3 + 1$

Answer: _____

1.6 Homework

Homework — Due next session

1. Classify and place on number line: -5 , 2.7 , $\sqrt{3}$, 0 , $11/4$

Answer: _____

2. Evaluate: $4 \times (6 - 2^2) + 3 \times |-5|$

Answer: _____

3. Convert 1.4 to a fraction.

Answer: _____

4. Give an example of each: rational non-integer, irrational, negative integer.

Answer: _____

5. If $|x| = 9$, what are the possible values of x ?

Answer: _____

6. Evaluate: $-3^2 + (-3)^2$ — explain the difference.

Answer: _____

7. Is the sum of two irrational numbers always irrational? Give an example or counterexample.

Answer: _____

8. Write $5/8$ as a decimal. Is it terminating or repeating?

Answer: _____

9. Apply distributive property: $7(2a - 3b + 1)$

Answer: _____

10. Find the value: $-|-14| + |7 - 20|$

Answer: _____

Chapter 2

Algebraic Expressions & Polynomials

Learning Objectives

- Identify terms, coefficients, and degrees of algebraic expressions
- Add, subtract, and multiply polynomials
- Factor polynomials using GCF, trinomial factoring, and difference of squares
- Apply algebraic identities

Key Formulas

- Degree of polynomial: highest power of the variable
- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $(a + b)(a - b) = a^2 - b^2$ (Difference of Squares)
- FOIL: $(a + b)(c + d) = ac + ad + bc + bd$
- GCF factoring: $ax + ay = a(x + y)$

2.1 Worked Examples

Example 1

Simplify: $3x^2 + 5x - 2x^2 + 7 - 3x$

Solution:

Combine like terms: $(3x^2 - 2x^2) + (5x - 3x) + 7 = x^2 + 2x + 7$

Example 2

Multiply: $(2x + 3)(x - 5)$

Solution:

FOIL: $= 2x \cdot x + 2x \cdot (-5) + 3 \cdot x + 3 \cdot (-5) = 2x^2 - 10x + 3x - 15 = 2x^2 - 7x - 15$

Example 3

Factor completely: $x^2 - 5x + 6$

Solution:

Find two numbers that multiply to 6 and add to -5: $(-2)(-3) = (x - 2)(x - 3)$

Example 4

Factor: $4x^2 - 25$

Solution:

Difference of squares: $a^2 - b^2 = (a+b)(a-b)$ $4x^2 - 25 = (2x)^2 - 5^2 = (2x + 5)(2x - 5)$

■ Common Mistakes

- $(a + b)^2 \neq a^2 + b^2$ — do not forget the middle term $2ab$!
- When subtracting polynomials, distribute the negative sign to ALL terms.

2.2 Practice

Independent Practice

1. Simplify: $5a^2 - 3a + 2a^2 + 6a - 9$

Answer: _____

2. Multiply: $(x + 4)(x + 3)$

Answer: _____

3. Multiply: $(3x - 1)(2x + 5)$

Answer: _____

4. Expand: $(x + 6)^2$

Answer: _____

5. Factor: $x^2 + 7x + 12$

Answer: _____

6. Factor: $x^2 - 16$

Answer: _____

7. Factor: $2x^2 - 8x$

Answer: _____

8. Factor: $x^2 - x - 12$

Answer: _____

2.3 Exit Ticket

Exit Ticket

1. What is the degree of: $4x^3 - 2x + 7$?

Answer: _____

2. Expand: $(x - 3)^2$

Answer: _____

3. Factor: $x^2 - 9$

Answer: _____

4. Simplify: $(2x^2 + 3x) - (x^2 - x + 4)$

Answer: _____

5. Factor out GCF: $6x^2y - 9xy^2$

Answer: _____

Chapter 3

Linear Equations & Inequalities**Learning Objectives**

- Solve one-variable linear equations and inequalities
- Solve word problems using linear equations
- Graph linear inequalities on a number line
- Understand compound inequalities (AND/OR)

Key Rules

- Addition/Subtraction Property of Equality: if $a = b$, then $a \pm c = b \pm c$
- Multiplication/Division Property of Equality: if $a = b$, then $ac = bc$
- FLIP the inequality sign when multiplying or dividing by a **NEGATIVE** number
- Compound AND: both conditions must be true simultaneously
- Compound OR: at least one condition must be true

3.1 Worked Examples**Example 1**

Solve: $3x - 7 = 14$

Solution:

$$3x - 7 = 14 \quad 3x = 21 \quad x = 7$$

Example 2

Solve: $2(x + 3) = 5x - 9$

Solution:

$$2x + 6 = 5x - 9 \quad 6 + 9 = 5x - 2x \quad 15 = 3x \quad x = 5$$

Example 3Solve and graph: $-2x + 4 > 10$ **Solution:**

$$-2x > 6 \text{ Divide by } -2 \text{ (flip sign): } x < -3 \text{ Graph: open circle at } -3, \text{ arrow pointing left}$$

Example 4

Word problem: A number decreased by 8 equals twice the number plus 3. Find the number.

Solution:

$$\text{Let } n = \text{the number} \quad n - 8 = 2n + 3 \quad -8 - 3 = 2n - n \quad n = -11$$

3.2 Practice

Independent Practice

1. Solve: $4x + 9 = 25$

Answer: _____

2. Solve: $5(x - 2) = 3x + 4$

Answer: _____

3. Solve: $x/3 - 2 = 4$

Answer: _____

4. Solve and graph: $3x - 1 \geq 8$

Answer: _____

5. Solve: $-5x > 25$

Answer: _____

6. A rectangle has perimeter 54. Its length is 3 more than twice its width. Find dimensions.

Answer: _____

7. Solve: $2 < 3x - 1 < 11$ (compound inequality)

Answer: _____

8. Solve: $|x - 2| = 5$

Answer: _____

3.3 Exit Ticket

Exit Ticket

1. Solve: $7x - 3 = 4x + 9$

Answer: _____

2. Solve: $-3x \leq 12$

Answer: _____

3. Is $x = -2$ a solution to $2x + 5 > 0$?

Answer: _____

4. Write an equation: five more than three times a number is 20.

Answer: _____

5. Solve: $|2x| = 8$

Answer: _____

Chapter 4

Coordinate Geometry

Learning Objectives

- Plot and read points on the Cartesian plane
- Calculate distance between two points and midpoint of a segment
- Find slope of a line and interpret its meaning
- Write equations of lines in slope-intercept and standard form

Key Formulas

- Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Midpoint: $M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$
- Slope: $m = (y_2 - y_1)/(x_2 - x_1)$
- Slope-intercept form: $y = mx + b$
- Standard form: $Ax + By = C$
- Point-slope form: $y - y_1 = m(x - x_1)$
- Parallel lines: same slope ($m_1 = m_2$)
- Perpendicular lines: slopes are negative reciprocals ($m_1 \times m_2 = -1$)

4.1 Worked Examples

Example 1

Find the distance between A(1, 2) and B(5, 5).

Solution:

$$d = \sqrt{((5-1)^2 + (5-2)^2)} = \sqrt{(16 + 9)} = \sqrt{25} = 5$$

Example 2

Find the midpoint of the segment joining P(-2, 4) and Q(6, -2).

Solution:

$$M = ((-2+6)/2, (4+(-2))/2) = (4/2, 2/2) = (2, 1)$$

Example 3

Write the equation of a line with slope 3 passing through (1, -2).

Solution:

$$y - (-2) = 3(x - 1) \quad y + 2 = 3x - 3 \quad y = 3x - 5$$

4.2 Practice

Independent Practice

1. Plot and label: A(3,4), B(-2,1), C(0,-3), D(-4,-2)

Answer: _____

2. Find distance between (0,0) and (3,4).

Answer: _____

3. Find midpoint of (-6, 2) and (4, -8).

Answer: _____

4. Find slope through (2, 5) and (6, 13).

Answer: _____

5. Write $y = mx + b$ for: slope = -2, y-intercept = 7

Answer: _____

6. Are lines $y = 3x + 1$ and $y = 3x - 4$ parallel, perpendicular, or neither?

Answer: _____

7. Write the equation of the line through (2,3) and (4,7).

Answer: _____

8. Find the x-intercept and y-intercept of $2x + 3y = 12$.

Answer: _____

Chapter 5

Introduction to Statistics

Learning Objectives

- Organize data using frequency tables, bar graphs, and histograms
- Calculate measures of central tendency: mean, median, mode
- Calculate range and identify outliers
- Interpret data from graphs and tables

Key Formulas

- Mean = Sum of all values $\div$ Number of values
- Median = middle value when data is ordered (average of two middle values if even count)
- Mode = most frequently occurring value
- Range = Maximum – Minimum
- Outlier: a value significantly higher or lower than the rest of the data

5.1 Worked Examples

Example 1

Find the mean, median, mode, and range of: 4, 7, 2, 9, 7, 5, 3

Solution:

Ordered: 2, 3, 4, 5, 7, 7, 9 Mean = $(2+3+4+5+7+7+9)/7 = 37/7 \approx 5.3$ Median = 5 (middle value) Mode = 7 (appears twice) Range = $9 - 2 = 7$

Example 2

Class test scores: 55, 72, 78, 80, 82, 85, 88, 90, 92, 95. Find the mean and identify any outliers.

Solution:

Mean = $(55+72+78+80+82+85+88+90+92+95)/10 = 817/10 = 81.7$ Outlier: 55 (significantly below the rest)

5.2 Practice

Independent Practice

1. Data: 12, 15, 18, 12, 20, 17, 15, 12. Find mean, median, mode, range.

Answer: _____

2. Which measure of central tendency is most affected by outliers?

Answer: _____

3. Create a frequency table for: A, B, A, C, B, A, D, B, A, C.

Answer: _____

4. A student scores 70, 85, 90, 78 on four tests. What score is needed on the 5th test to average 82?

Answer: _____

5. Describe the shape of a histogram that has a very long right tail.

Answer: _____

5.3 Exit Ticket

Exit Ticket

1. Find the mean of: 10, 20, 30, 40, 50.

Answer: _____

2. What is the median of: 3, 7, 9, 12, 15?

Answer: _____

3. What is the mode of: 5, 5, 7, 8, 9, 5, 7?

Answer: _____

4. A set has range 20 and minimum 8. What is the maximum?

Answer: _____

5. If one value in a data set is extremely large, which average is most affected?

Answer: _____

Answer Key — Selected Problems

Ch.1 Guided #2: Evaluate: $2 \times (3 + 4^2) - 10 \div 2 = 2 \times 19 - 5 = 38 - 5 = 33$

Ch.1 Indep. #2: $|-12| - |4-9| + 3^2 = 12 - 5 + 9 = 16$

Ch.2 Indep. #1: $5a^2 - 3a + 2a^2 + 6a - 9 = 7a^2 + 3a - 9$

Ch.2 Indep. #5: $x^2 + 7x + 12 = (x + 3)(x + 4)$

Ch.3 Indep. #1: $4x + 9 = 25 \rightarrow 4x = 16 \rightarrow x = 4$

Ch.3 Indep. #5: $-5x > 25 \rightarrow x < -5$

Ch.4 Indep. #2: $d = \sqrt{(9 + 16)} = \sqrt{25} = 5$

Ch.4 Indep. #5: $y = -2x + 7$

Ch.5 Indep. #1: Mean = 15.1, Median = 15, Mode = 12, Range = 8

Ch.5 Indep. #4: Needed: $5 \times 82 - (70+85+90+78) = 410 - 323 = 87$